

Generation Interconnection System Impact Study Report

For Queue Project Q15-0469

N. Shannon 138 kV

May 22, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0469. The Interconnection Customer proposes a 73.7 MW solar photovoltaic facility interconnecting at the N. Shannon 138 kV substation in the Kentucky area. The request was received on April 29, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$13,798,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BECKER-DISMAL RIVER 345.0 kV	FISHKILL PLAINS-DILLARD-ORANGE 345.0 kV	109.1	1,250	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DISMAL RIV-BECKER 345.0 kV	Reconductor line	\$132,932,000	10% (\$13,798,000)
	Total allocated to project		\$13,798,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.